



DEPARTMENT OF
COMPUTER SCIENCE



ALGORITHMS I

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LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- Explain what algorithms are and their relationship with computers.
- Outline the development process of algorithms.
- Discuss how to represent algorithms.



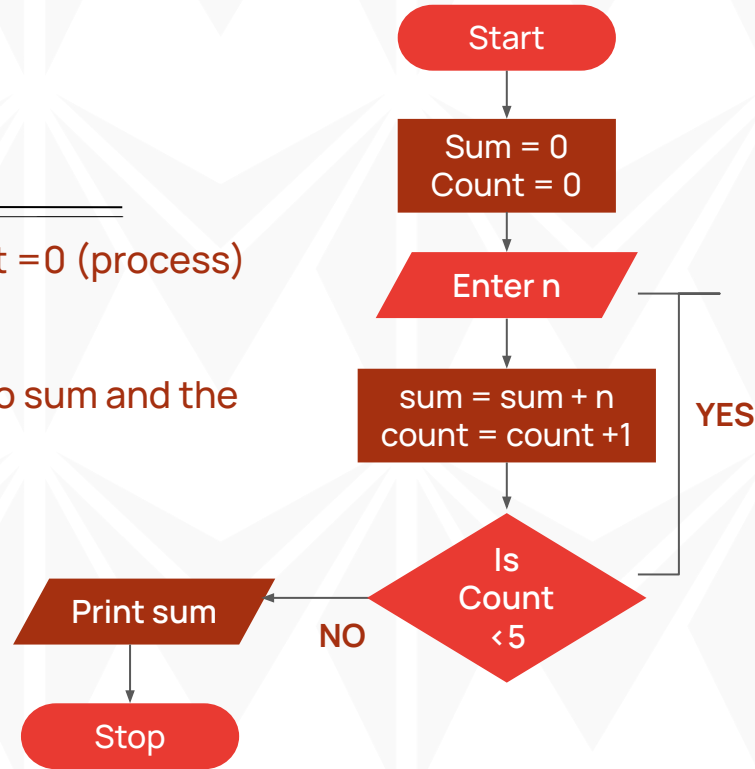
ALGORITHMS

- Algorithms are sets of commands that the computer can follow to carry out calculations or other problem-solving tasks.
- Algorithms are instructions implemented in a particular order to carry out a task.
- Algorithms consists of inputs, a process, and outputs.
- Algorithms can be represented as either pseudocodes or flowcharts.



ALGORITHMS: EXAMPLE

- Initialize sum = 0 and count = 0 (process)
- Enter n (I/O)
- Find sum + n and assign it to sum and the increment count by 1 (PROCESS)
- Is count < 5 (DECISION)
 - If YES go to step 2
 - Else
 - Print sum (I/O)





COMPUTERS AND ALGORITHMS

COMPUTER

- ❖ Computers are tools used to carry out a problem-solving plan.
- ❖ Computer programs are sets of instructions that describe the steps the computer has to follow in order to carry out a specific task or plan.



COMPUTERS AND ALGORITHMS

ALGORITHMS

- ❖ Algorithms are plans for solving problems.
- ❖ Algorithms are designed by people.
- ❖ Algorithms must be converted into a computer program.



ALGORITHMS DEVELOPMENT PROCESS

Algorithm development is a key step in the problem-solving process.

The algorithm development steps are;

- **Step 1:** Problem description
- **Step 2:** Problem analysis
- **Step 3:** High-level algorithm development
- **Step 4:** Algorithm refinement
- **Step 5:** Algorithm review



STEP 1: PROBLEM DESCRIPTION

Obtaining problem descriptions might be more complicated than it looks because, most times, there are two different parties involved in the process; the client (finding a solution) and the programmer (providing the solution).

Issues that can arise here are;

- Descriptions relying on unstated assumptions
- Ambiguous description
- Incomplete description



STEP 2: PROBLEM ANALYSIS

After describing the problem adequately, the problem needs to be analysed so that the starting and ending points of the problem-solving process is clear.

Questions to answer to decide on a starting point include;

- What data do we have?
- Where is the data?
- What formulas are related to the problem?
- What rules are required to manipulate the data?
- How are the data values related?



STEP 2: PROBLEM ANALYSIS

Questions to answer to decide on an ending point (when we have a solution) include;

- What results are we expecting?
- What parameters have changed in the items?
- What are the changes that have been made?
- What items are now non-existent?



MID-LESSON QUESTION

What are algorithms?



STEP 3: HIGH-LEVEL ALGORITHM DEVELOPMENT

This step involves the development of a high-level algorithm that is focused on the main parts of the solution and not on every minor detail.

The minor details can be added in subsequent development steps. It is better to develop a high-level algorithm first because it can be time wasting and costly to focus on every minor detail before, at least, coming up with a solution.



STEP 3: HIGH-LEVEL ALGORITHM DEVELOPMENT: EXAMPLE

A typical high-level algorithm for a meeting scheduling task is given below;

Step 1: Identify who needs to be at the meeting.

Step 2: Find common empty spaces in calendars.

Step 3: Choose one and tell everyone.

Step 4: Receive confirmation that everyone is still available.

Step 5: If response to 4 is negative, iterate from 2

ELSE IF

response to 4 is positive, identify meeting location.



STEP 4: ALGORITHM REFINEMENT

After writing a high-level algorithm that does not contain every minor detail, the next step is to provide more details about each high-level algorithm step (stepwise refinement) based on;

- Who will implement the algorithm?
- How much knowledge does the person that will implement the algorithm have?
- What are the computer specifications?



STEP 4: ALGORITHM REFINEMENT

We can refine one of the high-level algorithms in the previous example as follows;

Step 2: Find common empty spaces in calendars:

- Ask all to communicate availability or suggest times and get responses.
- Examine the response either manually or automatically.
- Find common openings, if any.
- Else, iterate with different time blocks or suggestions.



STEP 5: ALGORITHM REVIEW

After writing and refining the algorithm, it needs to be reviewed to confirm if the algorithm solves the problem. Some questions can also be asked at this stage;

- Can the algorithm only solve a particular problem or be applied to more general problems?
- Can the complexity of the algorithm be reduced?



ALGORITHM REPRESENTATION

It is important to present algorithms in ways that make them easily translatable into various programming languages. To achieve this, the algorithms should be represented in a form that has;

- Sequence
- Selection
- Repetition



ALGORITHM REPRESENTATION: SEQUENCE

The algorithm should be ordered in a step by step manner such that no step is skipped; unless there is a compelling reason to skip a step.

Step 1

Step 2

Step 3

Step 4

Step 5



ALGORITHM REPRESENTATION: SELECTION

There are instances where some steps can be selectively executed if some conditions are satisfied. The alternative steps to execute in this instances can be either one or more.

If condition is true
Then
perform activity



If condition is true
Then
perform activity 1
else
perform activity 2



ALGORITHM REPRESENTATION: REPETITION

There are also instances where the same steps need to be executed repeatedly. Since writing the same step on different lines is not sustainable, other methods can be explored to ensure that the same step is executed repeatedly.

```
while condition is true  
    do  
        perform activity
```



FURTHER READING RESOURCES

- Chaudhuri, A. B. (2020). Flowchart and algorithm basics: The art of programming. Mercury Learning and Information.
- Chaudhuri, A. B. (2005). The Art of Programming Through Flowcharts & Algorithms. Firewall Media.
- Edwards, S., Dorn, B., and Sanders (2012), D. Objects First: A brief introduction to object-oriented programming using Java to build simple Android applications.



REFERENCES

1. Chaudhuri, A. B. (2020). *Flowchart and algorithm basics: The art of programming*. Mercury Learning and Information.
2. Chaudhuri, A. B. (2005). *The Art of Programming Through Flowcharts & Algorithms*. Firewall Media.



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THANK YOU